

Requirements Engineering: Functional and Non-Functional Requirements

Problem Statement #17: Vaccination Cohort & Dose Scheduling System (Healthcare & Telemedicine)

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1. System summary

A public health administration platform that organises vaccination rollouts by cohort, enforces the minimum number of days between doses, schedules slots at vaccination centres, and issues QR vaccination certificates that can be verified.

Actors: Citizen Registrant, Vaccination Officer.

2. Where these requirements come from

Set	IDs	Source
Baseline	FR-001 to FR-005, NFR-001, NFR-002	Lab 1 requirement set, as tracked in Jira during Lab 2 (Kanban, Scrum and bug tracker)
Added in Lab 3	FR-006, FR-007, NFR-003 to NFR-005	Found while designing the architecture. Not yet in the Jira backlog

FR-001 and NFR-001 are the two sample requirements given in the problem statement.

3. Functional requirements

ID	Priority	Description	Acceptance criteria
FR-001	High	The system shall enforce dose interval rules (e.g., minimum 28 days after Dose 1) before unlocking Dose 2 booking for a citizen.	Pass: Dose 2 booking is blocked if the interval is under 28 days. Fail: An early vaccination slot is confirmed.
FR-002	High	The system shall allow a Citizen Registrant to create a profile with demographic details (age, comorbidity flags, occupation category) and automatically assign them to an eligible vaccination cohort/phase.	Pass: The registrant is auto-tagged with the correct cohort based on age/comorbidity rules. Fail: A registrant is placed in an ineligible cohort.
FR-003	High	The system shall let a Citizen Registrant search nearby vaccination centres by pincode/date and book an available time slot within that centre's daily capacity.	Pass: Booking succeeds only while booked count < centre capacity, and the slot count decrements. Fail: A centre accepts a booking beyond its capacity.
FR-004	Medium	The system shall allow a Vaccination Officer to check in a citizen, record the administered dose (vaccine brand, batch/lot number, date), and update the citizen's vaccination status.	Pass: Citizen status moves from "Slot Booked" to "Dose Administered" with a batch number stored. Fail: Status can be updated without a batch number.
FR-005	Medium	Upon completion of the required dose count, the system shall auto-generate a digitally signed QR vaccination certificate for the citizen, downloadable as a PDF.	Pass: The certificate QR encodes citizen ID, dose dates and a digital signature. Fail: A certificate is generated before the final dose is administered.
FR-006	High	The system shall let a Vaccination Officer scan a certificate QR code and see whether the certificate is genuine, together with the citizen's name and dose dates.	Pass: A genuine certificate shows "Valid" with matching details, and a certificate with any changed character shows "Invalid". Fail: A tampered certificate is shown as valid.
FR-007	Medium	The system shall send the citizen a confirmation when a slot is booked and a reminder 24 hours before the appointment (SMS or email).	Pass: Exactly one confirmation is queued for every successful booking, and one reminder 24 hours before the slot. Fail: A booking produces no confirmation, or produces duplicates.

4. Non-functional requirements

ID	Type	Priority	Description	Acceptance criteria
NFR-001	Performance & Security	High	The vaccination certificate verification endpoint shall authenticate digitally signed QR codes offline/online in under 150 ms.	Pass: Benchmarking tests confirm the target latency and security standards under simulated peak load.
NFR-002	Security & Privacy	High	All citizen personal and health data (name, ID, vaccination history) shall be encrypted at rest and in transit (TLS 1.2+/AES-256) and accessible only to authorized Vaccination Officers.	Pass: A security review confirms encryption at rest/in transit and role-based access control. Fail: PII is retrievable without authentication or found in plaintext.
NFR-003	Performance & Scalability	High	The system shall support at least 5,000 concurrent slot-booking requests during a cohort rollout window, with 95% of booking responses within 3 seconds.	Pass: A load test with 5,000 concurrent virtual users shows 95th-percentile response time of 3 s or less and zero double-bookings. Fail: The 95th percentile exceeds 3 s, or any slot is double-booked.
NFR-004	Availability	Medium	Dose recording and certificate verification shall remain usable when the booking function is overloaded or down.	Pass: With the booking service stopped in a test environment, an officer can still record a dose and verify a certificate. Fail: Either action fails because booking is unavailable.
NFR-005	Auditability	Medium	Every dose record shall store the officer ID and the date, and any later change to a record shall be logged.	Pass: Each dose record shows who recorded it and when, and an edit adds an audit entry. Fail: A record exists with no officer ID, or is changed with no trace.

5. Quality check of the requirement set

Property	How the set meets it
Unambiguous	Every requirement names one actor and one action
Testable	Every requirement has a pass condition that can be observed, most with a number (28 days, 150 ms, 5,000 users, 3 s)
Traceable	Each ID is linked to a use case, a component, a work package and a test in <code>RTM_Table.md</code>
Consistent	The seven baseline requirements are the same ones tracked on the Jira boards
Feasible	Each requirement maps to one owning service in the architecture

6. Coverage summary

	Baseline	Added	Total
Functional	5	2	7
Non-functional	2	3	5
Total	7	5	12

Requirements Traceability Matrix (RTM)

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1. Purpose

The RTM links every requirement forward to the place it is designed, planned, built and tested, and back to the use case it came from. It answers two questions quickly: "is every requirement covered?" and "if this requirement changes, what else must change?"

Column	Where it points
Use case	Lab 1 use-case diagram (UC-01 to UC-09)
Jira story	Scrum project Scrum_BPS#17 (SBPS-1 to SBPS-7), same stories as the Kanban project
Component	Lab 3 component diagram in 2-Architectural_Diagram
WBS	Work packages in Work_Breakdown_Structure.md
Test	Unit tests in test_vaccination_core_engine.py
Jira bug	Bug tracker project Bug_BPS#17 (BBPS-1 to BBPS-7)

2. Traceability matrix

Req.	Requirement (short)	Use case	Jira story	Component	WBS	Test	Jira bug	Status
FR-001	Dose interval lock (28 days)	UC-03	SBPS-1	Vaccination Record Service (IDoseEligibility)	4.2	TC-01, TC-02	BBPS-1	Built, unit tested
FR-002	Registration and cohort assignment	UC-01	SBPS-2	Citizen Registry Service	4.1	TC-03, TC-04, TC-05	BBPS-2	Built, unit tested
FR-003	Centre search and slot booking	UC-02	SBPS-3	Slot Booking Service	4.3	TC-06, TC-07, TC-08	BBPS-3	Built, unit tested
FR-004	Check-in and dose recording	UC-04, UC-05	SBPS-4	Vaccination Record Service, Officer Console	4.4	TC-10, TC-11	BBPS-4	Built, unit tested
FR-005	Signed QR certificate after final dose	UC-06	SBPS-5	Certificate Service	4.5	TC-12, TC-13	BBPS-5	Built, unit tested (PDF export not built)
FR-006	Certificate verification by QR scan	UC-07	not in backlog	Certificate Service, Officer Console	4.6	TC-14	none	Built, unit tested
FR-007	Booking confirmation and reminder	UC-08	not in backlog	Notification Service	4.8	TC-09	none	Confirmation built and tested, reminder not built
NFR-001	Verification under 150 ms	UC-07	SBPS-6	Certificate Service	4.6	TC-15 (local timing only)	BBPS-6	Peak-load benchmark planned (WBS 5.3)

Req.	Requirement (short)	Use case	Jira story	Component	WBS	Test	Jira bug	Status
NFR-002	Encryption and officer-only access	all	SBPS-7	API Gateway, one database per service	4.7	none yet	BBPS-7	Designed, security review planned (WBS 5.3)
NFR-003	5,000 concurrent bookings, p95 within 3 s	UC-02	not in backlog	Slot Booking Service (scaled out)	5.3	none yet	none	Designed, load test planned
NFR-004	Recording and verification survive a booking outage	UC-04, UC-07	not in backlog	Separate services and databases	5.3	none yet	none	Designed, failover test planned
NFR-005	Officer ID and date on every dose record	UC-05	not in backlog	Vaccination Record Service, Records DB	4.4	none yet	none	Officer ID and date stored, change log not built

UC-09 (Configure Cohort & Interval Rules) supplies the rule values that FR-001 and FR-002 enforce.

3. Jira bug traceability

Status is the Jira status at the time of the Lab 2 bug report. "Regression test" is the unit test in this repository that would fail if the bug came back.

Jira key	Bug	Requirement	Severity	Jira status	Regression test
BBPS-1	Dose 2 booking not blocked at exactly 27-day interval	FR-001	Critical	Done	TC-01
BBPS-2	Comorbidity flag ignored during cohort assignment	FR-002	High	In Progress	TC-04
BBPS-3	Centre slot allows double-booking at zero remaining capacity	FR-003	Critical	To Do	TC-07
BBPS-4	Dose record can be saved with an empty batch number	FR-004	Medium	To Do	TC-10
BBPS-5	QR certificate generated before final dose is administered	FR-005	Critical	In Progress	TC-12
BBPS-6	Certificate verification exceeds 150 ms target under peak load	NFR-001	High	To Do	TC-15 (partial: checks key reuse, not peak load)
BBPS-7	Citizen health data visible in plaintext in network logs	NFR-002	Critical	To Do	none (deployment setting, not engine code)

4. Coverage summary

Measure	Count
Requirements in the matrix	12 of 12
Requirements linked to a use case	12 of 12
Requirements linked to a component	12 of 12
Requirements with at least one automated test	8 of 12
Requirements with a Jira story	7 of 12 (the 5 added in Lab 3 are not in the backlog yet)
Jira bugs with a regression test	6 of 7

Measure	Count
Automated unit tests in the core engine	15, all passing

The four requirements without an automated test (NFR-002 to NFR-005) need a deployed system to check. They are planned under work package 5.3.